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Ryan J. Caverly[1]
Department of Aerospace Engineering & Mechanics,
University of Minnesota,
Minneapolis, MN 55455
e-mail: rcaverly@umn.edu

Sze Kwan Cheah
Department of Aerospace Engineering & Mechanics,
University of Minnesota,
Minneapolis, MN 55455
e-mail: cheah013@umn.edu

Keegan R. Bunker
Department of Aerospace Engineering & Mechanics,
University of Minnesota,
Minneapolis, MN 55455
e-mail: bunke029@umn.edu

Samir Patel
Department of Aerospace Engineering & Mechanics,
University of Minnesota,
Minneapolis, MN 55455
e-mail: samir004@umn.edu

Niko Sexton
Department of Aerospace Engineering & Mechanics,
University of Minnesota,
Minneapolis, MN 55455
e-mail: sexto164@umn.edu

Vinh L. Nguyen
Department of Aerospace Engineering & Mechanics,
University of Minnesota,
Minneapolis, MN 55455
e-mail: nguy2876@umn.edu

Online Self-Calibration of Cable-Driven Parallel Robots Using Covariance-Based Data Quality Assessment Metrics

This article presents an algorithm to perform self-calibration of cable-driven parallel robots (CDPRs), where the CDPR's end-effector pose is estimated in conjunction with the calibration of biases in CDPR's measurements. Two new metrics, known as the position dilution of precision (PDOP) and orientation dilution of precision (ODOP), are introduced as a means to quantify the quality of data collected with regards to self-calibration. These metrics are based on a covariance matrix that is computed online as part of the proposed self-calibration algorithm, which results in the PDOP and ODOP directly corresponding to the standard deviation of the position and orientation errors, respectively. These metrics are used to intuitively select which data points contribute to improved calibration, resulting in a computationally efficient algorithm requiring few data points to maintain accurate calibration. In addition, the PDOP and ODOP provide a means to assess when sufficient calibration data have been collected. Numerical results involving an inverse kinematic simulation with rigid cables and a dynamic simulation with flexible cables indicate that the proposed algorithm is capable of performing self-calibration in a computationally efficient manner. Moreover, the simulation results indicate that the proposed PDOP and ODOP metrics result in smaller position and orientation errors when used to prune the dataset compared to the observability indices found in the literature. Accuracy of the proposed algorithm is also confirmed through experiments when compared to ground-truth pose data.
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1 Introduction

Cable-driven parallel robots (CDPRs) are a class of robotic systems that make use of cables arranged in a parallel fashion to manipulate an end-effector or payload. Unique advantages of CDPRs over other classes of robotic systems include a large translational workspace, a high payload-to-weight ratio, and large acceleration capabilities. This has enabled intriguing CDPR applications, such as aerial videography of sporting events, motion simulation platforms [1], warehousing robots [2], and construction robots [3]. Successfully operating a CDPR requires accurate knowledge of the end-effector's pose (position and orientation). Forward kinematics, also known as direct kinematics, is typically used to compute the end-effector pose of a CDPR given measurements of the lengths of each of the CDPR's cables. The nonlinear nature of the loop-closure equations used to solve forward kinematics makes this a challenging procedure. A number of forward kinematics algorithms have been developed for CDPRs, including iterative [4], interval [5], hybrid [6], and machine-learning-based [7,8] formulations. An overview of the challenges when solving the forward kinematics of CDPRs and a survey of different algorithms can be found in Ref. [9, Ch. 4]. More recently, there have been

[1]Corresponding author.

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attempts to quantify the estimation error associated with forward kinematics using statistics and estimation theory [10,11], as well as sensitivity analyses [12–14]. Nonetheless, an accurate forward kinematics solution relies on accurate knowledge of the CDPR's geometry (e.g., the cable's attachment points on the end-effector and the outer structure) and knowledge of the initial cable lengths when using typical incremental winch encoder measurements. These quantities are difficult to obtain with precision in practice, which significantly affects the accuracy of the forward kinematics solution.

One option to improve the accuracy of forward kinematics is to include information from addition sensors within a sensor fusion algorithm [15]. Many different sensors have been proposed for this purpose, including exteroceptive vision systems [16,17], end-effector-mounted cameras [18], end-effector-mounted inertial measurement units (IMUs) [19–23], cable-orientation sensors [24,25], and direct cable-length sensors [26]. Although including additional sensors can improve accuracy of the end-effector's pose estimate, it does not address the fundamental issue that cable-length measurements typically have length bias errors when using incremental encoders and the CDPR geometry is never perfectly known. In certain situations, it is even possible that additional sensors will introduce new measurement biases (e.g., the biases associated with IMU measurements [22]).

Kinematic calibration is widely used within the robotics community to account for uncertainties present in the computations involved in forward kinematics [27,28]. For CDPRs, cable-length bias errors or other geometric parameters can be estimated by introducing additional sensors, such as a camera or vision-based referencing [29,30], force sensors [31], inclinometers mounted on the end-effector [32,33], laser displacement sensors [33], and color markers on the cables [34]. Although the information from these sensor can help with calibration, they require additional hardware and, which may be costly and not feasible in certain applications. Self-calibration algorithms make use of available winch encoder measurements to combine forward kinematics and calibration into a single operation. Self-calibration algorithms have been proposed for overconstrained CDPRs [35,36] and suspended CDPRs [37] with cable-length biases, as well as for CDPRs with cable-length biases and pulley kinematics [38,39]. Generally, the parameters to be estimated during calibration cannot be found with data from a single time-step or pose, and therefore, calibration data across multiple time-steps is required. As noted in Ref. [35], the number of data points used in this self-calibration process and the richness of motion/geometry present in these data points has a major impact on the accuracy of the calibration. A systematic approach to optimally select the data points used as part of self-calibration would allow for accurate calibration without requiring inordinate amounts of data collection and computation time. Observability indices have been proposed for this purpose [40–45], which involve computing the singular values of the Jacobian that maps either the parameters to be identified (e.g., cable-length biases) to the calibration poses [40–43] or the Jacobian that maps sensor noise to the calibration poses [44,45]. A number of variations on these observability indices exist, which have been compared to each other in different robotic applications [40–42]. Observability indices have been shown to be useful in choosing an optimal sequence of offline poses for the purpose of calibrating a 4 degrees-of-freedom (DOF) parallel robot [46]. Although these observability indices provide some quantification of the richness in the calibration data, they do not provide any direct insight into the amount of pose estimation error incurred during the calibration process. In other words, observability indices provide a relative indication of calibration data richness without quantifying absolute real-world improvements.

This article introduces two new data quality assessment indices inspired by global positioning system (GPS) navigation [47], known as the position dilution of precision (PDOP) and the orientation dilution of precision (ODOP). These metrics are computed based on the covariance of the self-calibration pose estimation error and can be directly related to the expected variance in the end-effector's position and orientation estimation error following self-calibration for a given calibration dataset. The PDOP and ODOP metrics are key components of a self-calibration algorithm proposed in this article in two different ways: (1) they provide a means to systematically select a small number of data points to include in the self-calibration process without sacrificing accuracy and (2) they provide an intuitive quantification of when sufficient rich data have been obtained to begin performing self-calibration. Their first use as a means of systematic data selection is analogous to how observability indices have been used in the literature; however, it is shown in Sec. 4.2.1 that using PDOP and ODOP for this purpose results in reduced pose estimation error. The use of PDOP and ODOP to intuitively assess whether sufficiently rich data have been collected is unique and can greatly improve the performance of a self-calibration algorithm, especially in an online implementation when computation time is limited. It is shown in Sec. 4.2.1 that performing self-calibration with insufficiently rich data can result in large pose estimation transients, which can lead to CDPR failures in practice. Due to their intuitive interpretation as the variance in the position and orientation errors, threshold values of PDOP and ODOP can easily be set to determine when sufficient calibration has been obtained.

The contributions of this work are the proposed PDOP and ODOP data quality assessment metrics, along with a self-calibration algorithm that makes use of these new metrics for data selection and assessment. The proposed algorithm features an inner loop that solves forward kinematics with the current calibration values and an outer loop that performs self-calibration. The work presented in this article is an extension of the self-calibration algorithm presented in the preliminary work of Ref. [48]. The additional contributions of this article in relation to Ref. [48] include direct comparisons between the proposed PDOP and ODOP metrics relative to observability indices from the literature, as well as an entirely new experimental assessment of the proposed algorithm.

The remainder of this article proceeds as follows. Section 2 introduces important notation and a mathematical description of the forward kinematics and self-calibration problem considered in this work. The proposed self-calibration algorithm is then presented in a general manner in Sec. 3 to accommodate CDPRs with different geometries and DOFs, a variety of sensors (e.g., cable orientation sensors [24] or direct cable-length sensors [26]), and/or generalized loop-closure equations (e.g., geometrico-static equations [37]). A numerical example of the algorithm applied to an overconstrained 6-DOF CDPR of the proposed algorithm are presented in Sec. 4, including inverse kinematics-based simulations with ideal cables and dynamics-based simulations with flexible cables. Section 5 includes experimental results with the proposed algorithm, followed by concluding remarks in Sec. 6.

2 Preliminaries

This section presents important notation, as well as a description of the general CDPR system considered in this article and the definition of the problem addressed in this work.

2.1 Notation. In this article, matrices are denoted by bold variables. The identity matrix of dimension $n \times n$ and the zero matrix of dimension $n \times m$ are written as $\mathbf{1}_{n\times n}$ and $\mathbf{0}_{n\times m}$, respectively. A reference frame, $\mathcal{F}_i$, is defined by three dextral (right-handed), orthonormal basis vectors, $\vec{i}^1$, $\vec{i}^2$, $\vec{i}^3$. The subscript $(\cdot)_k$ denotes a variable evaluated at time-step k, the superscript $(\cdot)^i$ denotes a variable concerning the CDPR's ith cable, the superscript $(\cdot)^{(j)}$ denotes a variable evaluated at the jth step within an iterative algorithm, and $\hat{(\cdot)}$ denotes the estimate of a variable. The continuous random variable $\mathbf{v} \sim \mathcal{N}(\bar{\mathbf{v}}, \mathbf{Q}_v)$ satisfies a normal distribution with mean $\bar{\mathbf{v}}$ and covariance $\mathbf{Q}_v = \mathbb{E}[(\mathbf{v}-\bar{\mathbf{v}})(\mathbf{v}-\bar{\mathbf{v}})^{\mathrm{T}}]$, where $\mathbf{Q}_v = \mathbf{Q}_v^{\mathrm{T}} \geq 0$ is a symmetric positive semidefinite matrix.

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2.2 System Description and Problem Definition. Consider an n-DOF CDPR with m cables, an example of which is shown in Fig. 1. The reference frame $\mathcal{F}_o$ is an inertial frame with point o at its origin. The position of the end-effector's center of mass relative to o is $\vec{r}$, and its components resolved in $\mathcal{F}_o$ are given by $\mathbf{r} \in \mathbb{R}^3$. The position of the ith cable's anchoring point (e.g., routing pulley or winch drum) relative to o is $\vec{a}^i$ and its components resolved in $\mathcal{F}_o$ are given by $\mathbf{a}^i$. The position of the ith cable's attachment point on the end-effector relative to its anchoring point is $\vec{\ell}^i$ and its components resolved in $\mathcal{F}_o$ are given by $\boldsymbol{\ell}^i$. In the case of a CDPR with a rigid-body end-effector, $\mathcal{F}_p$ denotes an end-effector-fixed frame, the position of the ith cable's attachment point on the end-effector relative to the end-effector center of mass is $\vec{b}^i$, and its components resolved in $\mathcal{F}_p$ are given by $\mathbf{b}^i$. In general, the pose of the end-effector is described by $\boldsymbol{\rho} \in \mathbb{R}^n$. In the case of a 6-DOF CDPR, $\boldsymbol{\rho}^{\mathrm{T}} = \begin{bmatrix} \mathbf{r}^{\mathrm{T}} & \boldsymbol{\theta}^{\mathrm{T}} \end{bmatrix}$, where $\boldsymbol{\theta} \in \mathbb{R}^3$ represents a parameterization of the rotation matrix $\mathbf{R}_{po}(\boldsymbol{\theta})$ or direction cosine matrix $\mathbf{C}_{po}(\boldsymbol{\theta})$ that describes the orientation of $\mathcal{F}_p$ relative to $\mathcal{F}_o$ (e.g., Euler angles). The direction cosine matrix and the rotation matrix are related by $\mathbf{C}_{po}(\boldsymbol{\theta}) = \mathbf{R}_{po}^{\mathrm{T}}(\boldsymbol{\theta})$.

It is assumed that a relationship between the CDPR's measurements and its pose is known to satisfy the generalized model

$$\mathbf{f}(\boldsymbol{\rho}, \mathbf{y}, \mathbf{v}, \boldsymbol{\beta}) = \mathbf{0}_{p\times 1} \tag{1}$$

where $\boldsymbol{\rho} \in \mathbb{R}^n$ contains the pose parameters, $\mathbf{y} \in \mathbb{R}^p$ contains linearly independent CDPR measurements, $\mathbf{v} \in \mathbb{R}^p$ is zero-mean Gaussian white noise that satisfies $\mathbf{v} \sim \mathcal{N}(\mathbf{0}_{p\times 1}, \mathbf{Q}_v)$ with $\mathbf{Q}_v \in \mathbb{R}^{p\times p}$, and $\boldsymbol{\beta} \in \mathbb{R}^q$ contains constant bias terms in the measurements (e.g., cable-length measurement biases, errors in the position of the end-effector cable attachment points, or errors in the position of cable-routing pulleys). The dimension p of the generalized model (i.e., the number of measurements and equations in Eq. (1)) varies depending on the modeling assumptions used and the sensor information available. As an example, in the case of a typical forward kinematics formulation, Eq. (1) represents the $p = m$ loop-closure equations, $\mathbf{y} \in \mathbb{R}^m$ contains the cable-length measurements, and $\boldsymbol{\beta} \in \mathbb{R}^m$ represents the biases in the cable-length measurements, which may be due to initial calibration errors or unknown initial lengths. Details on this particular application and how the loop-closure equations relate to Eq. (1) are provided in Sec. 4.1. More generally, Eq. (1) can represent geometrico-static equations with $\mathbf{y}$ containing cable-length and cable tension measurements (e.g., Ref. [37]) or it can represent loop-closure equations augmented with additional measurements (e.g., the inclusion of cable orientation sensors [24] or direct cable-length measurement sensors [26]), in which case $p > m$. It is assumed that $n \le p < n + q$, which implies that the biases $\boldsymbol{\beta}$ is not observable from the measurements at a given time-step (i.e., it is not possible to simultaneously estimate $\boldsymbol{\rho}$ and $\boldsymbol{\beta}$ given a single time-step of data). It is also worth noting that although in practice the measurement noise $\mathbf{v}$ may not be exactly Gaussian, it is conventional in many applications to approximate or overbound noise's distribution with a Gaussian distribution [49]. The overarching goal of this paper is to develop an online self-calibration method that accurately estimates both $\boldsymbol{\rho}$ and $\boldsymbol{\beta}$ without requiring large amounts of data or excessive computation. The formulation of the proposed method presented in Sec. 3 is kept general to account for all possible cases of (1), with only the assumptions that all measurements contained in $\mathbf{y}$ are available at each time-step and the function $\mathbf{f}(\boldsymbol{\rho}, \mathbf{y}, \mathbf{v}, \boldsymbol{\beta})$ is differentiable with respect to $\boldsymbol{\rho}$, $\mathbf{v}$, and $\boldsymbol{\beta}$.

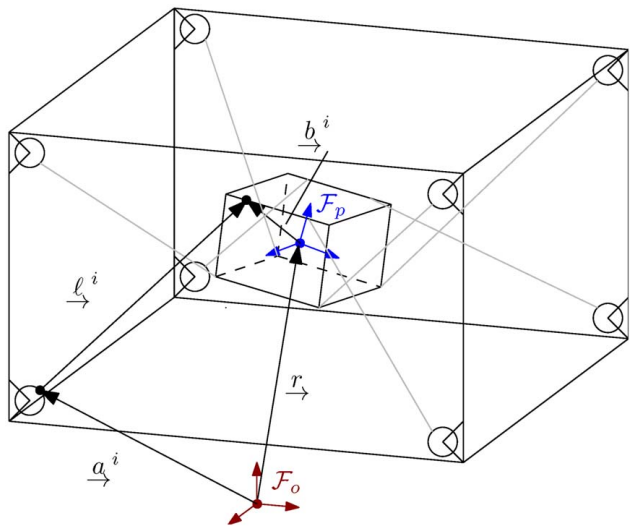


Fig. 1 An example of an overconstrained 6-DOF CDPR

3 Online Forward Kinematics and Self-Calibration Formulation

The proposed forward kinematics and online self-calibration algorithm features an inner-loop and outer-loop structure. The inner loop estimates the end-effector pose $\hat{\boldsymbol{\rho}}_k$ at each time-step using the current best knowledge of the bias term $\hat{\boldsymbol{\beta}}$. The algorithm's outer loop makes use of measurements across N time-steps to estimate the bias and uses least-squares estimation theory and newly defined PDOP and ODOP metrics to automatically determine when this bias estimate is accurate enough for implementation in the inner loop.

Both the inner and outer loops of the algorithm involve linearizing (1) about $\boldsymbol{\rho} = \bar{\boldsymbol{\rho}}$, $\mathbf{v} = \mathbb{E}[\mathbf{v}] = \mathbf{0}_{p\times 1}$, and $\boldsymbol{\beta} = \bar{\boldsymbol{\beta}}$, which results in

$$\begin{aligned}\mathbf{f}(\boldsymbol{\rho}, \mathbf{y}, \mathbf{v}, \boldsymbol{\beta}) \approx{}& \mathbf{f}(\bar{\boldsymbol{\rho}}, \mathbf{y}, \mathbf{0}_{p\times 1}, \bar{\boldsymbol{\beta}}) + \mathbf{J}_\rho(\bar{\boldsymbol{\rho}}, \mathbf{y}, \bar{\boldsymbol{\beta}})\delta\boldsymbol{\rho} \\ &+ \mathbf{J}_v(\bar{\boldsymbol{\rho}}, \mathbf{y}, \bar{\boldsymbol{\beta}})\mathbf{v} + \mathbf{J}_\beta(\bar{\boldsymbol{\rho}}, \mathbf{y}, \bar{\boldsymbol{\beta}})\delta\boldsymbol{\beta}\end{aligned} \tag{2}$$

where

$$\mathbf{J}_\rho(\bar{\boldsymbol{\rho}}, \mathbf{y}, \bar{\boldsymbol{\beta}}) = \left.\frac{\partial \mathbf{f}}{\partial \boldsymbol{\rho}}\right|_{\bar{\boldsymbol{\rho}}, \mathbf{y}, \mathbf{0}_{p\times 1}, \bar{\boldsymbol{\beta}}}$$

$$\mathbf{J}_v(\bar{\boldsymbol{\rho}}, \mathbf{y}, \bar{\boldsymbol{\beta}}) = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{v}}\right|_{\bar{\boldsymbol{\rho}}, \mathbf{y}, \mathbf{0}_{p\times 1}, \bar{\boldsymbol{\beta}}}$$

$$\mathbf{J}_\beta(\bar{\boldsymbol{\rho}}, \mathbf{y}, \bar{\boldsymbol{\beta}}) = \left.\frac{\partial \mathbf{f}}{\partial \boldsymbol{\beta}}\right|_{\bar{\boldsymbol{\rho}}, \mathbf{y}, \mathbf{0}_{p\times 1}, \bar{\boldsymbol{\beta}}}$$

The inner-loop and outer-loop portions of the proposed algorithm are presented in the following subsections, followed by a comparison between the proposed PDOP and ODOP metrics and the observability indices found in the literature.

3.1 Inner-Loop Forward Kinematics. At each time-step k, the inner loop of the algorithm assumes a fixed estimate of $\hat{\boldsymbol{\beta}}$ is available and solves the forward kinematics as a weighted nonlinear least-squares optimization problem

$$\min_{\hat{\boldsymbol{\rho}}_k} \mathbf{f}^{\mathrm{T}}(\hat{\boldsymbol{\rho}}_k, \mathbf{y}_k, \mathbf{0}_{p\times 1}, \hat{\boldsymbol{\beta}})\mathbf{H}^{-1}(\hat{\boldsymbol{\rho}}_k, \mathbf{y}_k, \hat{\boldsymbol{\beta}})\mathbf{f}(\hat{\boldsymbol{\rho}}_k, \mathbf{y}_k, \mathbf{0}_{p\times 1}, \hat{\boldsymbol{\beta}}) \tag{3}$$

where $\mathbf{H}(\hat{\boldsymbol{\rho}}_k, \mathbf{y}_k, \hat{\boldsymbol{\beta}})$ is the covariance of the noise associated with $\mathbf{f}(\hat{\boldsymbol{\rho}}_k, \mathbf{y}_k, \mathbf{0}_{p\times 1}, \hat{\boldsymbol{\beta}})$ and is calculated as $\mathbf{H}(\hat{\boldsymbol{\rho}}_k, \mathbf{y}_k, \hat{\boldsymbol{\beta}}) = \mathbf{J}_v(\hat{\boldsymbol{\rho}}_k, \mathbf{y}_k, \hat{\boldsymbol{\beta}})\mathbf{Q}_v\mathbf{J}_v^{\mathrm{T}}(\hat{\boldsymbol{\rho}}_k, \mathbf{y}_k, \hat{\boldsymbol{\beta}})$. The solution to Eq. (3) represents the best least unbiased estimate of $\hat{\boldsymbol{\rho}}_k$ [50, p. 84], which provides the pose estimate with the minimum variance in its error, assuming that $\hat{\boldsymbol{\beta}} \approx \boldsymbol{\beta}$. This same type of weighted nonlinear least-squares optimization problem was previously used to incorporate cable orientation sensors into a CDPR forward kinematics algorithm [24]. The weighted nonlinear least-squares problem is solved iteratively by initializing $\hat{\boldsymbol{\rho}}_k^{(0)}$ and solving for $\hat{\boldsymbol{\rho}}_k$ through the iterative

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Levenberg–Marquardt algorithm given by

$$\hat{\boldsymbol{\rho}}_k^{(j+1)} = \hat{\boldsymbol{\rho}}_k^{(j)} - \left(\hat{\mathbf{J}}_\rho^{(j)^{\mathrm{T}}} \hat{\mathbf{H}}^{(j)^{-1}} \hat{\mathbf{J}}_\rho^{(j)} + \eta \mathbf{1}_{n\times n}\right)^{-1} \hat{\mathbf{J}}_\rho^{(j)^{\mathrm{T}}} \hat{\mathbf{H}}^{(j)^{-1}} \hat{\mathbf{f}}^{(j)} \tag{4}$$

where $\hat{\mathbf{J}}_\rho^{(j)} = \mathbf{J}_\rho(\hat{\boldsymbol{\rho}}_k^{(j)}, \mathbf{y}_k, \hat{\boldsymbol{\beta}})$, $\hat{\mathbf{H}}^{(j)} = \mathbf{H}(\hat{\boldsymbol{\rho}}_k^{(j)}, \mathbf{y}_k, \hat{\boldsymbol{\beta}})$, $\hat{\mathbf{f}}^{(j)} = \mathbf{f}(\hat{\boldsymbol{\rho}}_k^{(j)}, \mathbf{y}_k, \mathbf{0}_{p\times 1}, \hat{\boldsymbol{\beta}})$, $\eta > 0$ provides numerical damping to the iterative calculations, and the iteration is terminated when a specified stopping criterion is satisfied. The covariance of the pose estimation error after j iterations at the kth time-step is given as follows [50, p. 84]:

$$\begin{aligned}\mathbf{P}_k^{(j)} &= \mathbb{E}[(\boldsymbol{\rho}_k - \hat{\boldsymbol{\rho}}_k^{(j)})(\boldsymbol{\rho}_k - \hat{\boldsymbol{\rho}}_k^{(j)})^{\mathrm{T}}] \\ &= \left(\mathbf{J}_\rho^{\mathrm{T}}(\hat{\boldsymbol{\rho}}_k^{(j)}, \mathbf{y}_k, \hat{\boldsymbol{\beta}}) \mathbf{H}^{-1}(\hat{\boldsymbol{\rho}}_k^{(j)}, \mathbf{y}_k, \hat{\boldsymbol{\beta}}) \mathbf{J}_\rho(\hat{\boldsymbol{\rho}}_k^{(j)}, \mathbf{y}_k, \hat{\boldsymbol{\beta}})\right)^{-1}\end{aligned} \tag{5}$$

3.2 Outer-Loop Bias Estimation. After accumulating N time-steps of measurements, the time indices of the outer loop estimator data set are initialized as $k_1 = 1$, $k_2 = 2$, …, $k_N = N$. These time indices are to be updated as part of the outer-loop self-calibration algorithm as new, richer data are obtained. The linearization in Eq. (2) associated with time indices $k_1, \ldots, k_N$ is performed about the nominal values of $\bar{\boldsymbol{\beta}}, \bar{\boldsymbol{\rho}}_{k_1}, \ldots \bar{\boldsymbol{\rho}}_{k_N}$, yielding

$$\begin{aligned}\bar{\mathbf{f}}(\mathbf{x}, \mathbf{y}, \mathbf{v}) \approx \bar{\mathbf{f}}(\bar{\mathbf{x}}, \mathbf{y}, \mathbf{0}_{pN\times 1}) + \bar{\mathbf{J}}_\rho(\bar{\mathbf{x}}, \mathbf{y}) \begin{bmatrix} \delta\boldsymbol{\rho}_{k_1} \\ \vdots \\ \delta\boldsymbol{\rho}_{k_N} \end{bmatrix} + \bar{\mathbf{J}}_v(\bar{\mathbf{x}}, \mathbf{y}) \begin{bmatrix} \mathbf{v}_{k_1} \\ \vdots \\ \mathbf{v}_{k_N} \end{bmatrix} \\ + \bar{\mathbf{J}}_\beta(\bar{\mathbf{x}}, \mathbf{y}) \delta\boldsymbol{\beta},\end{aligned}$$

where $\mathbf{x}^{\mathrm{T}} = \begin{bmatrix} \boldsymbol{\beta}^{\mathrm{T}} & \boldsymbol{\rho}_{k_1}^{\mathrm{T}} & \cdots & \boldsymbol{\rho}_{k_N}^{\mathrm{T}} \end{bmatrix}$, $\mathbf{y}^{\mathrm{T}} = \begin{bmatrix} \mathbf{y}_{k_1}^{\mathrm{T}} & \cdots & \mathbf{y}_{k_N}^{\mathrm{T}} \end{bmatrix}$, $\mathbf{v}^{\mathrm{T}} = \begin{bmatrix} \mathbf{v}_{k_1}^{\mathrm{T}} & \cdots & \mathbf{v}_{k_N}^{\mathrm{T}} \end{bmatrix}$,

$$\bar{\mathbf{f}}(\mathbf{x}, \mathbf{y}, \mathbf{v}) = \begin{bmatrix} \mathbf{f}(\boldsymbol{\rho}_{k_1}, \mathbf{y}_{k_1}, \mathbf{v}_{k_1}, \boldsymbol{\beta}) \\ \vdots \\ \mathbf{f}(\boldsymbol{\rho}_{k_N}, \mathbf{y}_{k_N}, \mathbf{v}_{k_N}, \boldsymbol{\beta}) \end{bmatrix},$$

$$\bar{\mathbf{J}}_\rho(\mathbf{x}, \mathbf{y}) = \mathrm{diag}\{\mathbf{J}_\rho(\boldsymbol{\rho}_{k_1}, \mathbf{y}_{k_1}, \boldsymbol{\beta}), \ldots, \mathbf{J}_\rho(\boldsymbol{\rho}_{k_N}, \mathbf{y}_{k_N}, \boldsymbol{\beta})\},$$

$$\bar{\mathbf{J}}_v(\mathbf{x}, \mathbf{y}) = \mathrm{diag}\{\mathbf{J}_v(\boldsymbol{\rho}_{k_1}, \mathbf{y}_{k_1}, \boldsymbol{\beta}), \ldots, \mathbf{J}_v(\boldsymbol{\rho}_{k_N}, \mathbf{y}_{k_N}, \boldsymbol{\beta})\},$$

$$\bar{\mathbf{J}}_\beta(\mathbf{x}, \mathbf{y}) = \begin{bmatrix} \mathbf{J}_\beta(\boldsymbol{\rho}_{k_1}, \mathbf{y}_{k_1}, \boldsymbol{\beta}) \\ \vdots \\ \mathbf{J}_\beta(\boldsymbol{\rho}_{k_N}, \mathbf{y}_{k_N}, \boldsymbol{\beta}) \end{bmatrix}$$

The covariance of the estimation error in the bias and the end-effector poses across these N time-steps is then computed as follows:

$$\bar{\mathbf{P}} = \mathbb{E}[(\mathbf{x} - \hat{\mathbf{x}})(\mathbf{x} - \hat{\mathbf{x}})^{\mathrm{T}}] = \left(\bar{\mathbf{J}}_{\beta\rho}^{\mathrm{T}}(\hat{\mathbf{x}}, \mathbf{y}) \bar{\mathbf{H}}^{-1}(\hat{\mathbf{x}}, \mathbf{y}) \bar{\mathbf{J}}_{\beta\rho}(\hat{\mathbf{x}}, \mathbf{y})\right)^{-1}$$

where

$$\bar{\mathbf{H}}(\mathbf{x}, \mathbf{y}) = \mathrm{diag}\{\mathbf{H}(\boldsymbol{\rho}_{k_1}, \mathbf{y}_{k_1}, \boldsymbol{\beta}), \ldots, \mathbf{H}(\boldsymbol{\rho}_{k_N}, \mathbf{y}_{k_N}, \boldsymbol{\beta})\},$$

$$\bar{\mathbf{J}}_{\beta\rho}(\mathbf{x}, \mathbf{y}) = \begin{bmatrix} \bar{\mathbf{J}}_\beta(\mathbf{x}, \mathbf{y}) & \bar{\mathbf{J}}_\rho(\mathbf{x}, \mathbf{y}) \end{bmatrix}$$

The metrics used to assess the quality of the data in these N time-steps of measurements are denoted as the position dilution of precision (PDOP) and the ODOP, which are defined as follows:

$$\begin{aligned}\mathrm{PDOP} &= \sqrt{\frac{1}{N}\sum_{i=1}^{N} \mathrm{trace}\left(\bar{\mathbf{P}}_{n(i-1)+q+1:\, n(i-1)+q+n_r}\right)} \\ &\approx \mathbb{E}\left[\|\mathbf{r}_k - \hat{\mathbf{r}}_k\|_2\right]\end{aligned} \tag{6}$$

$$\begin{aligned}\mathrm{ODOP} &= \sqrt{\frac{1}{N}\sum_{i=1}^{N} \mathrm{trace}\left(\bar{\mathbf{P}}_{n(i-1)+q+n_r+1:\, ni+q+1}\right)} \\ &\approx \mathbb{E}\left[\left\|\boldsymbol{\theta}_k - \hat{\boldsymbol{\theta}}_k\right\|_2\right]\end{aligned} \tag{7}$$

where $\bar{\mathbf{P}}_{i:j}$ denotes a submatrix of $\bar{\mathbf{P}}$ given by its i through j rows and columns, and $n_r \le n$ is the number of CDPR end-effector translational DOFs (e.g., $n_r = 3$ in the case of a 6-DOF CDPR with three translational DOFs). The terms PDOP and ODOP are inspired by similar terms used within the GPS community to describe the richness of data available when computing a navigation solution based on the geometry and number of visible satellites [47, p. 208]. Furthermore, the relationship between PDOP and the average position error is described within the application of GPS in [47, p. 208]. The ODOP metric is analogous to the attitude dilution of precision or Euler angle dilution of precision (EADOP) metrics used in certain aerospace navigation applications [51,52].

The values of PDOP and ODOP provide a way to quantify the quality or richness of the data collected with regards to self-calibration. Larger values of PDOP and ODOP indicate that either not enough data points are being used for this computation (i.e., N is too small) or the data points do not feature rich enough motion. Moreover, the values of PDOP and ODOP have physical significance, as the PDOP and ODOP are approximately equal to the standard deviation in the position and orientation error, respectively [47, p. 208]. The PDOP and ODOP are used for two purposes within the proposed self-calibration method. Their first use is that they provide a means to keep the amount of data points involved in the outer-loop computation low, with minimal sacrifice to accuracy. Specifically, after N time-steps of measurements are accumulated, the PDOP and ODOP values can be used to determine whether the current time-step of data provides additional richness to the existing N time-steps of data. In the proposed scheme, the PDOP and ODOP are calculated with the existing N time-steps of data and then later when the number of time-steps is greater than N, with the current time of data replacing a randomly selected time-step of data within the existing N time-steps. If the PDOP and ODOP are both smaller with the new time-step of data, then the data from the randomly selected time-step of data are removed from the outer-loop computation and the current time-step of data is added, otherwise the current time-step of data is not incorporated into the outer loop. The second use of the PDOP and ODOP is their ability to assess whether the outer-loop self-calibration has rich enough data to be able to trust the bias estimates it computes for use within the inner-loop forward kinematics calculations. Threshold values of PDOP and ODOP can be set to prevent the computation of the biases until specified accuracies in end-effector position and orientation are achievable. This prevents inaccurate bias estimates from being computed and implemented within the inner loop, which could lead to even worse forward kinematics pose estimates than without the use of bias estimates. This choice of thresholds is discussed further in the numerical example of Sec. 4.

If it is determined that both the PDOP and ODOP are lower than their threshold values, then $\hat{\mathbf{x}}$ is solved for through the weighted nonlinear least-squares optimization problem:

$$\min_{\hat{\mathbf{x}}} \bar{\mathbf{f}}^{\mathrm{T}}(\hat{\mathbf{x}}, \mathbf{y}, \mathbf{0}_{pN\times 1}) \bar{\mathbf{H}}^{-1}(\hat{\mathbf{x}}, \mathbf{y}) \bar{\mathbf{f}}(\hat{\mathbf{x}}, \mathbf{y}, \mathbf{0}_{pN\times 1})$$

Similar to the inner-loop forward kinematics optimization problem, the solution to this optimization problem represents the best least unbiased estimate of $\hat{\mathbf{x}}$ [50, p. 84]. This optimization problem is solved iteratively using the Levenberg–Marquardt update law:

$$\hat{\mathbf{x}}^{(j+1)} = \hat{\mathbf{x}}^{(j)} - \left(\hat{\mathbf{J}}_{\beta\rho}^{(j)^{\mathrm{T}}} \hat{\mathbf{H}}^{(j)^{-1}} \hat{\mathbf{J}}_{\beta\rho}^{(j)} + \eta \mathbf{1}_{(nN+q)\times(nN+q)}\right)^{-1} \hat{\mathbf{J}}_{\beta\rho}^{(j)^{\mathrm{T}}} \hat{\mathbf{H}}^{(j)^{-1}} \hat{\mathbf{f}}^{(j)} \tag{8}$$

where $\hat{\mathbf{J}}_{\beta\rho}^{(j)} = \bar{\mathbf{J}}_{\beta\rho}(\hat{\mathbf{x}}^{(j)}, \mathbf{y})$, $\hat{\mathbf{H}}^{(j)} = \bar{\mathbf{H}}(\hat{\mathbf{x}}^{(j)}, \mathbf{y})$, $\hat{\mathbf{f}}^{(j)} = \bar{\mathbf{f}}(\hat{\mathbf{x}}^{(j)}, \mathbf{y})$ and $\hat{\mathbf{x}}^{(0)}$ is initialized with the previous estimates of the bias and pose

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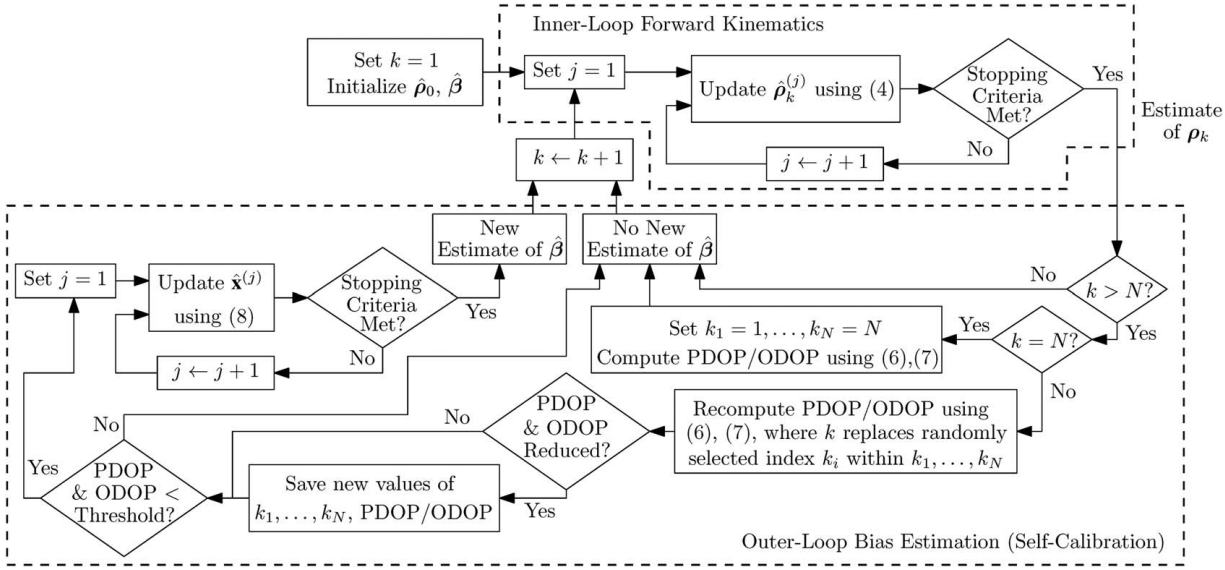


Fig. 2 Flowchart of the proposed self-calibration algorithm composed of an inner-loop for forward kinematics and an outer-loop for bias estimation

parameters. The bias estimate is then extracted from $\hat{\mathbf{x}}^{(j+1)}$ following the termination of this iteration and set as $\hat{\boldsymbol{\beta}}_k$, which is then used from the current time-step onward within the inner loop, until a new outer-loop computation is performed. A depiction of the overall self-calibration algorithm and its inner-loop/outer-loop architecture is provided in Fig. 2.

Although the proposed self-calibration method is described as an online strategy where the CDPR end-effector is following a dynamic trajectory, it is worth noting that all of the computations described can be performed using offline static trajectories, where the CDPR end-effector is moved to set of poses in the workspace. In this case, the PDOP and ODOP metrics could still be helpful in determining which set of poses result in a rich enough data set for calibration.

3.3 Comparison to Observability Indices. Observability indices have been discussed in the literature to serve a similar purpose as the proposed PDOP and ODOP metrics [40,42,45]. These indices involve computing the singular values of the Jacobian that maps either the biases to the CDPR pose (i.e., $\boldsymbol{\beta}$ to $\boldsymbol{\rho}$) or the measurements to the combined biases and calibration poses (i.e., $\mathbf{v}$ to $\mathbf{x}$). In the case of self-calibration, this involves computing the singular values of $\bar{\mathbf{J}}_{\beta\rho}(\hat{\mathbf{x}}, \mathbf{y})$. A few of the most commonly used observability indices are given by

$$O_1 = \sigma_{\min}$$
$$O_2 = \sigma_{\min}/\sigma_{\max}$$
$$O_3 = \sigma_{\min}^2/\sigma_{\max}$$
$$O_4 = \frac{1}{\frac{1}{\sigma_1} + \frac{1}{\sigma_2} + \cdots + \frac{1}{\sigma_h}}$$
$$O_5 = \frac{1}{\sqrt{m}}(\sigma_1 \sigma_2 \cdots \sigma_h)^{\frac{1}{h}}$$

where $\sigma_1, \sigma_2, \ldots, \sigma_h$ are all singular values of $\bar{\mathbf{J}}_{\beta\rho}(\hat{\mathbf{x}}, \mathbf{y})$, and $\sigma_{\min}$ and $\sigma_{\max}$ are the minimum and maximum singular values of $\bar{\mathbf{J}}_{\beta\rho}(\hat{\mathbf{x}}, \mathbf{y})$, respectively.

These observability indices are designed to provide a quantified measure of the richness in the data used for calibration and thus can be useful in determining which data points to use when performing calibration. Although they may provide a useful relative metric, unlike the PDOP and ODOP, these indices do not have units, which makes it difficult to intuitive gauge whether sufficiently rich data have been collected. A direct comparison between the proposed PDOP and ODOP and observability indices with regards to data selection is included in the numerical results of Sec. 4.2.1.

4 Numerical Results on a 6-DOF CDPR With Cable-Length Biases

This section presents the implementation of the self-calibration algorithm proposed in Sec. 3 for a 6-DOF over-constrained CDPR with biases in its cable-length measurements. First, Sec. 4.1 details the specifics of the proposed self-calibration algorithm for this application, followed by numerical simulation results in Sec. 4.2.

4.1 Forward Kinematics Formulation for a 6-DOF CDPR With Inelastic Cables and Cable-Length Biases. Consider a 6-DOF overconstrained CDPR with $m > n$ cables, where an incremental encoder on each winch is used to obtain cable-length measurements relative to an initial length. Assuming that the cables are inelastic and pulley kinematics can be ignored, the position vector of the ith cable's attachment point on the end-effector relative to its attachment is given by

$$\boldsymbol{\ell}^i(\boldsymbol{\rho}) = \mathbf{r} + \mathbf{C}_{po}^{\mathrm{T}}(\boldsymbol{\theta})\mathbf{b}^i - \mathbf{a}^i$$

where $\mathbf{a}^i$ is the position of the ith cable's anchoring point relative to the origin o resolved in $\mathcal{F}_o$ and $\mathbf{b}^i$ is the position of the ith cable's attachment point on the end-effector relative to its anchoring point resolved in $\mathcal{F}_o$. In the case of a CDPR with a rigid-body end-effector, $\mathcal{F}_p$ denotes an end-effector-fixed frame, the position of the ith cable's attachment point on the end-effector relative to the end-effector center of mass is $\vec{b}^i$, and its components resolved in $\mathcal{F}_p$ are given by $\mathbf{b}^i$. Each cable-length measurement is modeled to satisfy

$$y^i = \left\|\boldsymbol{\ell}^i(\boldsymbol{\rho})\right\|_2 + v^i + \beta^i \tag{9}$$

where y^i is the ith cable's length measurement, $v^i \in \mathcal{N}(0, \sigma^{i^2})$ is the noise associated with this measurement, and β^i is the cable-length calibration error bias associated with the ith cable. Defining the measurement Eq. (9) for all m cables and rewriting the relation in the form of

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(1) results in

$$\mathbf{0}_{m\times 1} = \mathbf{f}(\boldsymbol{\rho}, \mathbf{y}, \mathbf{v}, \boldsymbol{\beta}) = \mathbf{y} - \mathbf{g}(\boldsymbol{\rho}) - \mathbf{v} - \boldsymbol{\beta} \tag{10}$$

where $\mathbf{y} \in \mathbb{R}^m$ contains all the CDPR cable-length measurements, $\mathbf{g}^{\mathrm{T}}(\boldsymbol{\rho}) = [\|\boldsymbol{\ell}^1(\boldsymbol{\rho})\|_2 \ \dots \ \|\boldsymbol{\ell}^m(\boldsymbol{\rho})\|_2]$, $\mathbf{v}^{\mathrm{T}} = [v^1 \ \dots \ v^m]$ contains zero-mean Gaussian cable-length measurement noise with $\mathbf{v} \in \mathcal{N}(\mathbf{0}_{m\times 1}, \mathbf{Q}_v)$, and $\mathbf{Q}_v = \mathrm{diag}\{\sigma^{1^2}, \dots, \sigma^{m^2}\}$, while $\boldsymbol{\beta}^{\mathrm{T}} = [\beta^1 \ \dots \ \beta^m]$ contains the calibration error bias in the cable-length measurements.

For this specific application, the Jacobians included in the generic linearization of Eq. (2) are specified as $\mathbf{J}_\beta = \mathbf{J}_v = -\mathbf{1}_{m\times m}$, while $\mathbf{J}_\rho(\boldsymbol{\rho}) = -\frac{\partial \mathbf{g}}{\partial \boldsymbol{\rho}}$ is the Jacobian that maps pose rates to cable-length rates and its transpose maps cable forces to the end-effector wrench. If a 3-2-1 Euler angle sequence (also known as *zyx* Euler angles) is used to describe the end-effector's orientation (i.e., $\boldsymbol{\theta}^{\mathrm{T}} = \begin{bmatrix}\theta_1 & \theta_2 & \theta_3\end{bmatrix}$), then

$$\mathbf{J}_\rho(\boldsymbol{\rho}) = \begin{bmatrix} \frac{\boldsymbol{\ell}^{1^{\mathrm{T}}}(\boldsymbol{\rho})}{\|\boldsymbol{\ell}^1(\boldsymbol{\rho})\|_2}\begin{bmatrix}\mathbf{1}_{3\times 3} & -\mathbf{C}_{po}^{\mathrm{T}}(\boldsymbol{\theta})\mathbf{b}^{1^\times}\mathbf{S}(\boldsymbol{\theta})\end{bmatrix} \\ \vdots \\ \frac{\boldsymbol{\ell}^{m^{\mathrm{T}}}(\boldsymbol{\rho})}{\|\boldsymbol{\ell}^m(\boldsymbol{\rho})\|_2}\begin{bmatrix}\mathbf{1}_{3\times 3} & -\mathbf{C}_{po}^{\mathrm{T}}(\boldsymbol{\theta})\mathbf{b}^{m^\times}\mathbf{S}(\boldsymbol{\theta})\end{bmatrix} \end{bmatrix}$$

where $\mathbf{C}_{po}(\boldsymbol{\theta}) = \mathbf{C}_1(\theta_1)\mathbf{C}_2(\theta_2)\mathbf{C}_3(\theta_3)$; $\mathbf{C}_i(\theta)$, i = 1, 2, 3, are principal direction cosine matrices (*x*, *y*, and *z* Euler angle rotations) [53]:

$$\mathbf{S}(\boldsymbol{\theta}) = \begin{bmatrix} 1 & 0 & -\sin(\theta_2) \\ 0 & \cos(\theta_1) & \sin(\theta_1)\cos(\theta_2) \\ 0 & -\sin(\theta_1) & \cos(\theta_1)\cos(\theta_2) \end{bmatrix}$$

and $(\cdot)^\times$ represents the cross-product operator defined by

$$\mathbf{u}^\times = \begin{bmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{bmatrix}$$

with $\mathbf{u}^{\mathrm{T}} = \begin{bmatrix}u_1 & u_2 & u_3\end{bmatrix}$. Analytic expressions for $\mathbf{J}_\rho(\boldsymbol{\rho})$ with different parameterizations of the end-effector orientation, including constrained parameterizations, such as the quaternion, direction cosine matrix, or rotation matrix, are found in Refs. [10,11].

4.2 Numerical Simulations. Numerical simulations of a 6-DOF CDPR with eight cables and a geometry similar to that of IPAnema 2 described in Ref. [9, p. 319] are performed in this section. The dimensions of the CDPR, as shown in Fig. 3, are 1.43 m × 0.76 m × 0.93 m, with exact dimensions provided in Ref. [10]. Two sets of simulations are performed: inverse kinematics-based simulations with rigid cables and dynamics-based simulations with linear elastic cables. In both cases, a discrete time-step of $T = 0.005$ s is used and the desired pose trajectory of the end-effector is solved for using the kinematic equations:

$$\begin{aligned} \mathbf{r}_k &= \mathbf{r}_{k-1} + T\mathbf{v}_{k-1} \\ \mathbf{v}_k &= \mathbf{v}_{k-1} + T\dot{\mathbf{v}}_{k-1} \\ \mathbf{C}_{p_k o} &= \exp\left(-T\boldsymbol{\omega}_{k-1}^\times\right)\mathbf{C}_{p_{k-1}o} \end{aligned}$$

where $\mathbf{v}_k = \dot{\mathbf{r}}_k$,

$$\begin{aligned} \dot{\mathbf{v}}_k^{\mathrm{T}} &= -0.15\begin{bmatrix}\cos(kT) & \cos(1.5kT) & \sin(kT)\end{bmatrix}\ \mathrm{m/s^2} \\ \boldsymbol{\omega}_k^{\mathrm{T}} &= -4.5\begin{bmatrix}3\cos(1.25kT) & 3\cos(1.5kT) & \cos(kT)\end{bmatrix}\ \mathrm{deg/s} \end{aligned}$$

with initial conditions $\mathbf{r}_0^{\mathrm{T}} = \begin{bmatrix}0.15 & 0.15 & 0.465\end{bmatrix}$ m, $\mathbf{v}_0^{\mathrm{T}} = \begin{bmatrix}0.0003 & 0.0004 & 0.15\end{bmatrix}$ m/s,

$$\mathbf{C}_{p_0 o} = \begin{bmatrix} \cos(\theta_{3_0}) & \sin(\theta_{3_0}) & 0 \\ -\sin(\theta_{3_0}) & \cos(\theta_{3_0}) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and $\theta_{3_0} = 10$ deg. The resulting pose trajectory is shown in Fig. 4, where a 3-2-1 Euler-angle sequence is used to parameterize the orientation of the end-effector.

All implementations of the proposed algorithm assume loop-closure equations of the form in Eq. (10). The damping and stopping criteria tolerances are chosen as $\eta = 0.001$, $\|\hat{\boldsymbol{\rho}}_k^{(j+1)} - \hat{\boldsymbol{\rho}}_k^{(j)}\|_2 < 10^{-9}$, and $\|\hat{\mathbf{x}}_k^{(j+1)} - \hat{\mathbf{x}}_k^{(j)}\|_2 < 10^{-12}$. It is worth noting that the units of the norms $\|\hat{\boldsymbol{\rho}}_k^{(j+1)} - \hat{\boldsymbol{\rho}}_k^{(j)}\|_2$ and $\|\hat{\mathbf{x}}_k^{(j+1)} - \hat{\mathbf{x}}_k^{(j)}\|_2$ do not have a well-defined meaning as the units of $\hat{\boldsymbol{\rho}}_k$ and $\hat{\mathbf{x}}_k$ are not homogeneous. To resolve this issue, other more physically meaningful stopping criteria could be employed or a normalized version of these matrices could be used. We do not explore these options in this work, but will consider them in future studies. In all simulated cases, the initial pose estimate used in the Levenberg–Marquardt algorithm is the pose estimate from the previous time-step. The threshold values of PDOP and ODOP for the outer loop are chosen as 10 mm and 5 deg, respectively, although two other sets of threshold values are also compared in Sec. 4.2.1. The number of time-steps to include in the outer-loop computations is typically chosen as $N = 30$, although values of $N = 10$ and $N = 100$ are also tested in Sec. 4.2.1.

4.2.1 Inverse Kinematics-Based Simulations. Inverse kinematics-based simulations are performed, where zero-mean

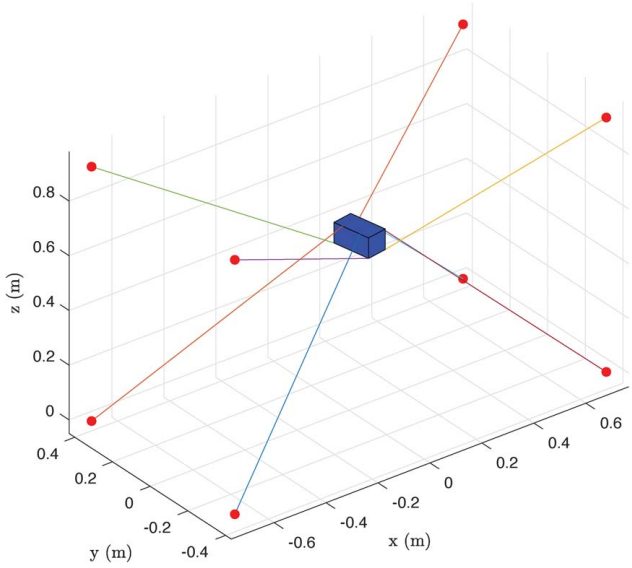


Fig. 3 Depiction of the 6-DOF over-constrained CDPR used in the numerical simulations of Sec. 4

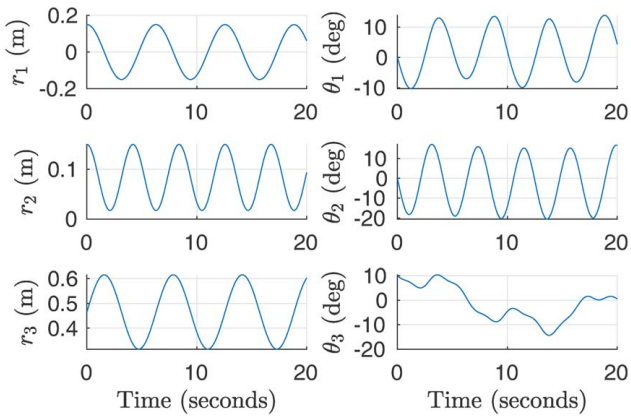


Fig. 4 Plots of the simulated desired end-effector pose trajectory versus time, where $\mathbf{r} = [r_1 \ r_2 \ r_3]^{\mathrm{T}}$ and $\boldsymbol{\theta} = [\theta_1 \ \theta_2 \ \theta_3]^{\mathrm{T}}$

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Gaussian measurement noise with standard deviation $\sigma_{\ell_i} = 0.5$ mm, $i = 1, \ldots, 8$ and arbitrarily-chosen constant cable-length biases $\boldsymbol{\beta} = \begin{bmatrix} 5 & 30 & -10 & -30 & -10 & 5 & -30 & 10 \end{bmatrix}^{\mathrm{T}}$ mm are added to the cable lengths before proceeding with forward kinematics.

The proposed algorithm is first implemented with the standard PDOP and ODOP thresholds, with the number of outer-loop time-steps varied between $N = 10$, $N = 30$, and $N = 100$. The pose estimation error versus time with $\pm 3\sigma$ bounds computed from the covariance matrix $\mathbf{P}_k$ in (5) are presented in Fig. 5. Specifically, the $\pm 3\sigma$ bounds on each component of the pose estimation error are calculated by taking the square root of the diagonal entries of the covariance matrix $\mathbf{P}_k$ at each time-step. The results in Fig. 5 demonstrate that similar levels of pose estimation accuracy are achieved with $N = 10$, $N = 30$, and $N = 100$. The main difference when varying the value of N is that it affects the amount of time it takes to meet the threshold values of PDOP and ODOP, as well as the time elapsed before performing the first outer-loop bias estimate computation. This emphasizes one of the main benefits of the proposed self-calibration algorithm, which is the reduction in computation time (e.g., the size of the matrices inverted can be reduced by a factor of 10 when using $N = 10$ instead of $N = 100$), with minimal effect on the estimation accuracy.

To emphasize the role that the threshold values of PDOP and ODOP play in the proposed algorithm, simulations are performed with $N = 30$ and three different threshold values: 50 mm and 30 deg, followed by 10 mm and 5 deg, and finally 5 mm and 3 deg. Plots of PDOP and ODOP, as well as $\|\mathbf{r} - \hat{\mathbf{r}}\|_2$ and $\|\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}\|_2$, versus time are presented in Fig. 6. The results demonstrate the

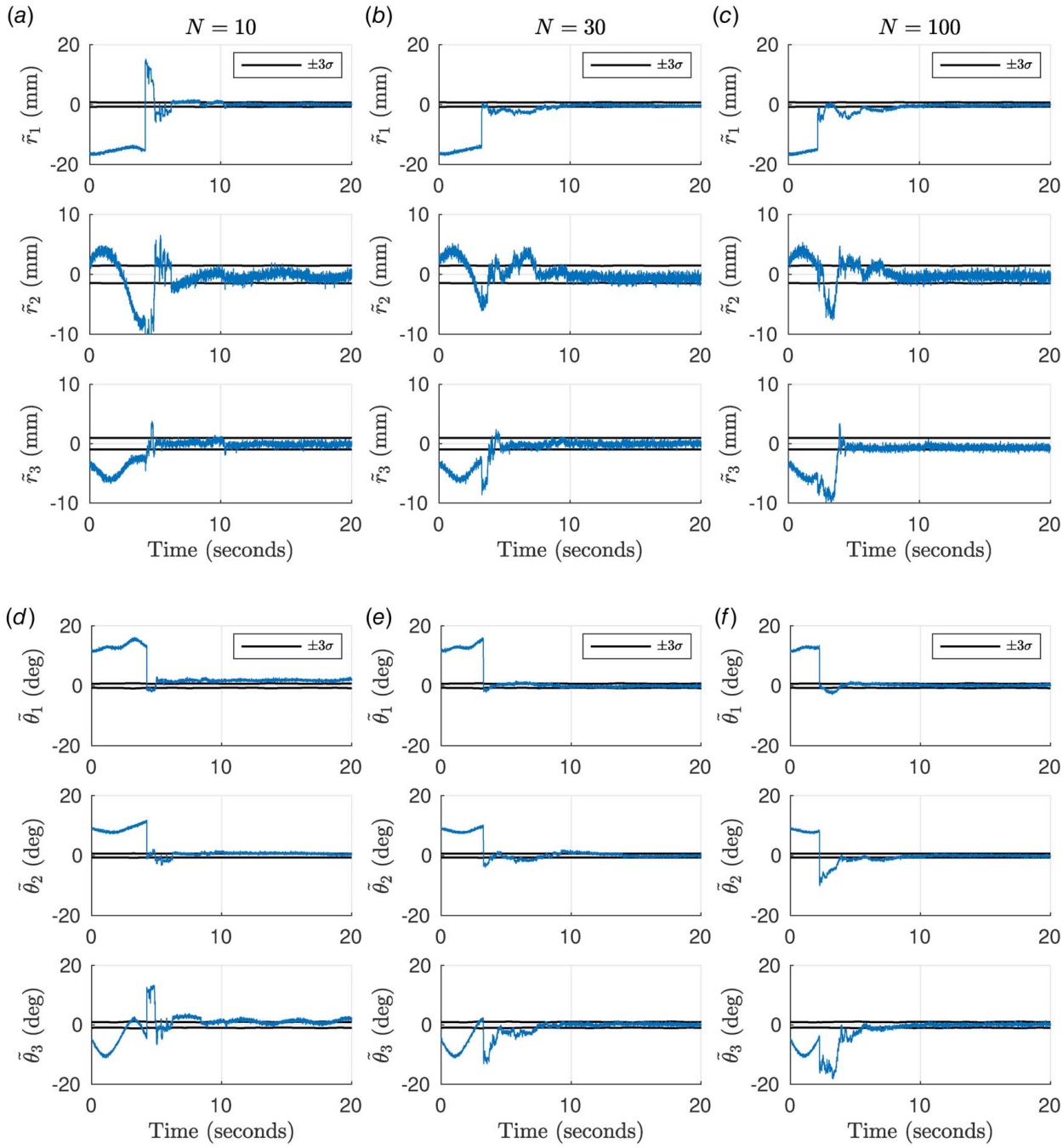


Fig. 5 Plots of the pose estimation error and $\pm 3\sigma$ bounds computed using (5) versus time for different choices of N, where $\mathbf{r} - \hat{\mathbf{r}} = [\tilde{r}_1 \;\; \tilde{r}_2 \;\; \tilde{r}_3]^{\mathrm{T}}$ and $\theta - \hat{\theta} = [\tilde{\theta}_1 \;\; \tilde{\theta}_2 \;\; \tilde{\theta}_3]^{\mathrm{T}}$, for $N = 10$ in (a) and (d), $N = 30$ in (b) and (e), and $N = 100$ in (c) and (f)

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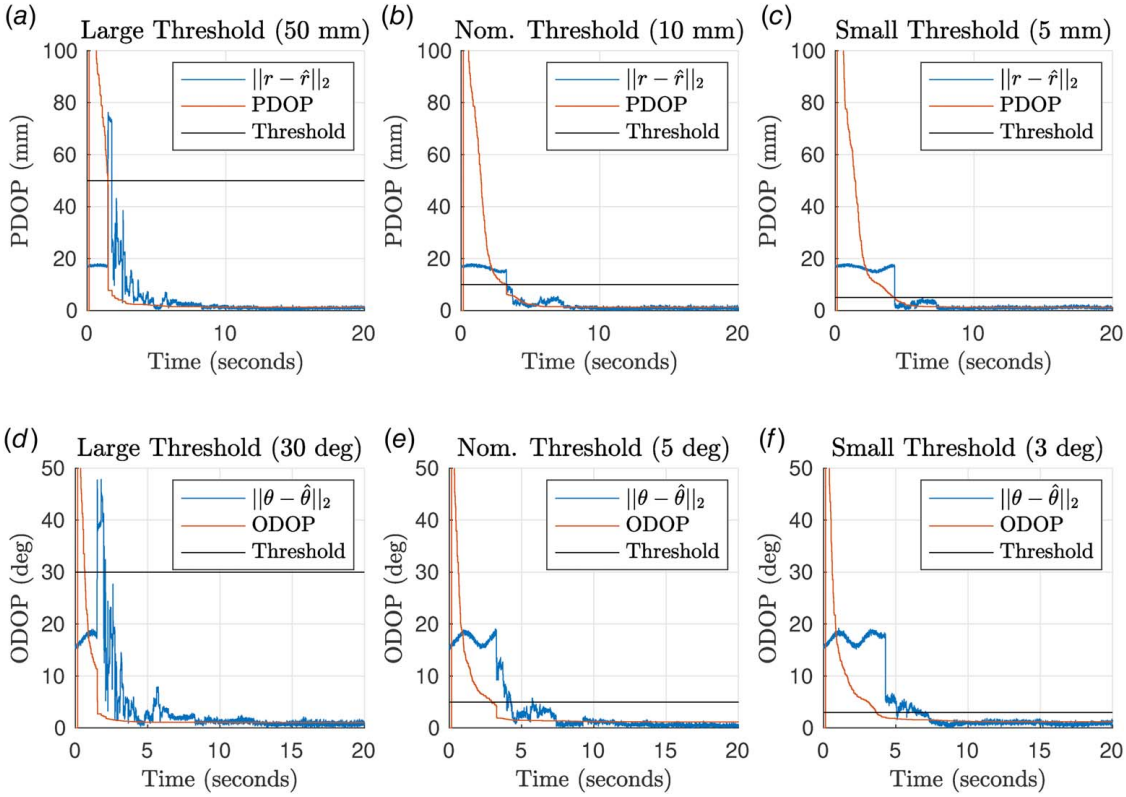


Fig. 6 Plots of the PDOP and ODOP, as well as the norm of the position and orientation estimation errors for different choices of PDOP and ODOP threshold values. Results are presented with *50* mm and *30* deg in (*a*) and (*d*), *10* mm and *5* deg in (*b*) and (*e*), and *5* mm and *3* deg in (*c*) and (*f*). The horizontal lines in the plots represents the threshold PDOP and ODOP values.

importance of collecting rich enough data (i.e., setting low enough PDOP and ODOP thresholds) before performing an outer-loop bias computation and implementing it within the inner-loop forward kinematics, otherwise large transient errors are possible (as shown in Figs. 6(*a*) and 6(*b*)). Figure 6 also demonstrates that the PDOP and ODOP values give accurate quantifications of the steady-state position and orientation estimation errors, respectively.

The random selection method used to determine which data points to include within the outer-loop self-calibration method is admittedly simple and seemingly suboptimal. However, it has been seen through simulation trials that not only is it a computationally efficient method but also it yields intuitive results and performs nearly optimally. Figure 7 includes the three-dimensional end-effector trajectory tested in simulation along with the data points on the trajectory that were selected by the proposed algorithm for the $N = 30$ case with PDOP and ODOP thresholds of 10 mm and 5 deg, respectively. It can be seen in this plot that the algorithm selects data points that are well spaced out in the workspace, which intuitively increases the richness in the data. To test the optimality of our random selection approach, simulation runs are performed, where at each time-step a complete search across all $N = 30$ stored calibration data points is done to find the optimal data point to replace with the current data point. The meaning of "optimal" in these tests is somewhat ambiguous, as there turns out to be a tradeoff between minimizing the PDOP and the ODOP. To this end, the objective function PDOP + αODOP is minimized, where each test increments α within the range $\alpha \in [10^{-5}, 10^{3}]$. The PDOP and ODOP values obtained at the end of these simulations are plotted in Fig. 8, where it is apparent that a Pareto curve is formed. Figure 8 also includes the PDOP and ODOP values from the end of the simulation with the proposed random replacement strategy, which are impressively close to the Pareto curve from the optimized data selection simulations. This demonstrates the effectiveness of the seemingly naive random replacement strategy, while significantly reducing the computation time by a factor of N (i.e., only one computation is needed versus checking all N possibilities).

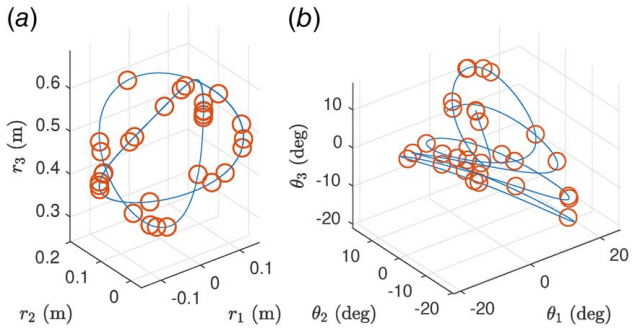


Fig. 7 Three-dimensional plots of the nominal (*a*) position and (*b*) orientation trajectory with circles visualizing the $N = 30$ data points used for bias estimation at the end of the trajectory from in the simulation results presented in Figs. 5(*a*) and 5(*b*)

To provide a comparison to the observability indices found in the literature, Monte Carlo simulations are performed, first using PDOP and ODOP to determine whether a new data are to be included within the self-calibration procedure, then using the observability indices described in Sec. 3.3 for this purpose. Monte Carlo simulations of 100 runs are performed with each method using $N = 30$, thresholds of PDOP =10 mm and ODOP =5 deg, along with all of the same parameters used in this section's numerical simulations. When PDOP and ODOP are used as determining factors to decide

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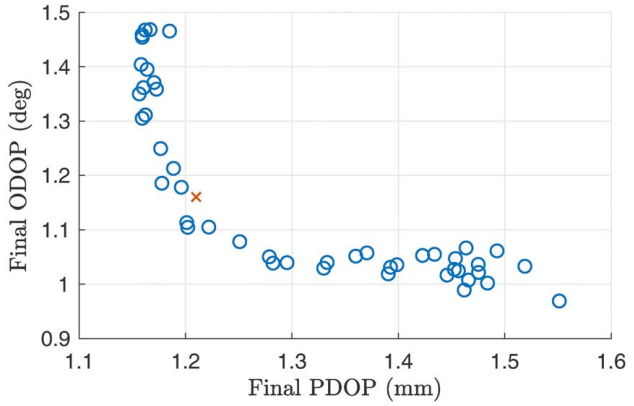


Fig. 8 Plot of the PDOP and ODOP at the end of the simulation trajectories, where the calibration data is chosen by minimizing $PDOP + \alpha ODOP$ for α within the range $\alpha \in [10^{-5}, 10^{3}]$. All of these simulations consider $N = 30$, and PDOP and ODOP thresholds of *10* mm and *5* deg, respectively. The final PDOP and ODOP values from the same simulation case with the proposed method that includes random replacement of the calibration data are labeled with an "x" marker.

whether to include a data point within the calibration set, the values of PDOP and ODOP must both decrease with the new data point for it to be included. With the observability indices, the observability index must increase with the inclusion of the new data point in order for it to be included. Results of the Monte Carlo simulations are shown in Fig. 9, where plots of the average steady-state position and orientation error computed across the last 5 s of the simulation are given. The results in Fig. 9 demonstrate that although all of the implementations lead to reasonable error levels, the use of PDOP and ODOP as a selection criteria results in the least average error.

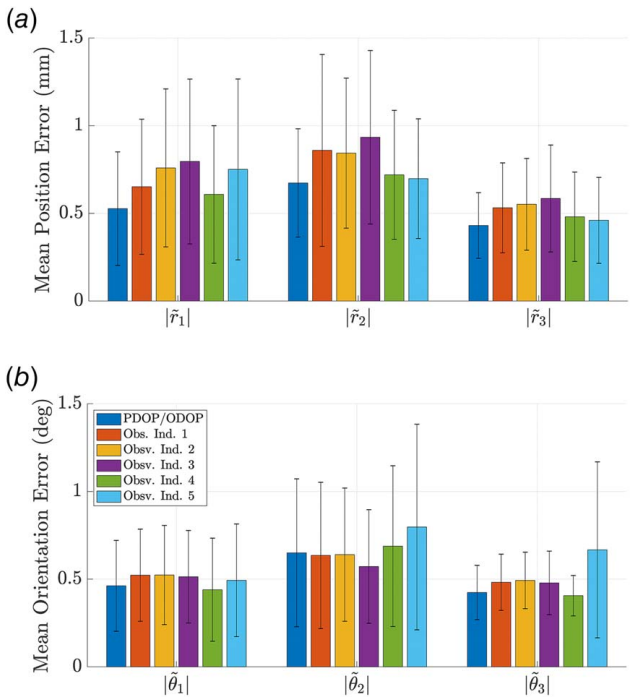


Fig. 9 Bar plots of the (*a*) mean position error and (*b*) mean orientation error when using either the PDOP and ODOP or observability indices to select the data points used for self-calibration, where $\mathbf{r} - \hat{\mathbf{r}} = [\tilde{r}_1 \ \tilde{r}_2 \ \tilde{r}_3]^T$ and $\theta - \hat{\theta} = [\tilde{\theta}_1 \ \tilde{\theta}_2 \ \tilde{\theta}_3]^T$. The results include 100 Monte Carlo runs, and the bars indicate the average mean error across all runs, while the error bars indicate the standard deviation across all runs.

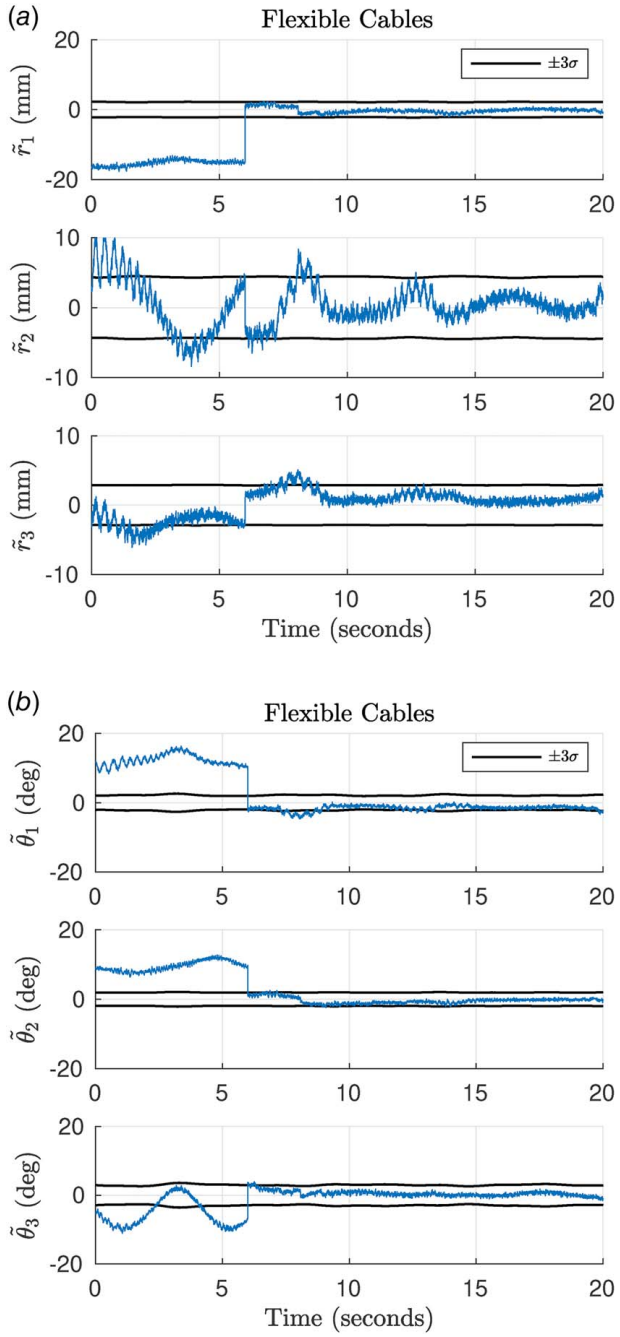


Fig. 10 Plots of the (*a*) position and (*b*) orientation estimation errors, along with the $\pm 3\sigma$ bounds computed using Eq. (5) versus time with the dynamics-based simulation with linear elastic cables, where $\mathbf{r} - \hat{\mathbf{r}} = [\tilde{r}_1 \ \tilde{r}_2 \ \tilde{r}_3]^T$ and $\theta - \hat{\theta} = [\tilde{\theta}_1 \ \tilde{\theta}_2 \ \tilde{\theta}_3]^T$

This provides additional confidence that the proposed PDOP and ODOP metrics are useful in selecting a rich set of data points for self-calibration.

4.2.2 Dynamics-Based Simulations With Linear Elastic Cables. To demonstrate robustness of the proposed forward kinematics and self-calibration algorithm, dynamics-based simulations are performed that make use of the linear elastic cable model in Ref. [54]. This model includes second-order motor dynamics of the form

$$J_i \ddot{\theta}_i + C_i \dot{\theta}_i + r_i \theta_i f_i = \tau_i$$

where θ_i is the angle of the ith motor, $J_i = 3 \times 10^{-4}$ kg·m^2 is the inertia of the ith motor, $C_i = 1 \times 10^{-5}$ N·m·s is the damping

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Fig. 11 The 6-DOF CDPR with eight cables used in the experimental results of Sec. 5. The CDPR's cables are highlighted for improved visualization.

coefficient of the ith motor, $r_i = 12.7$ mm is the radius of the ith winch, and f_i is the tension of the ith cable. The input τ_i is the torque provided to the ith motor, which is generated to track the desired trajectory using the closed-loop control method in Ref. [54], with a proportional gain of $K_p = 1$ N·m and a derivative gain of $K_d = 0.6$ N·m·s. The cable force f_i is computed using the linear elastic model

$$f_i = \frac{\mathrm{EA}}{\ell_{o,i}}(\ell_i - \ell_{0,i})$$

where EA = 10 kN is the stiffness of the cable, ℓ_i is the instantaneous length of the ith cable, and $\ell_{o,i}$ is the original length of the ith cable. The end-effector payload has a mass of 1 kg, and its second moment of mass resolved in $\mathcal{F}_p$ is $\mathbf{J}_p = 10^{-2}\mathrm{diag}\{2.81, 1.13, 2.81\}$ kg·m^2. Further details on the dynamic model are found in Ref. [54].

The proposed algorithm is implemented with no a priori knowledge of the vibrations in the cables (i.e., the same algorithm and equations presented in Sec. 4.1 are used in this example). To account for this, the algorithm assumes that the zero-mean Gaussian white noise in the cable length measurements is artificially increased to a standard deviation of 1.5 mm. To be clear, the actual zero-mean Gaussian white noise added to the cable lengths measurements remains the same as shown in Sec. 4.2.1; however, the algorithm uses a covariance of nine times $\mathbf{Q}_v$ to indirectly account for cable vibrations. Results of the pose estimation errors versus time are included in Fig. 10, which shows a marked improvement in the pose estimation error after the outer-loop self-calibration begins at around 6 s. The $\pm 3\sigma$ bounds computed from the covariance matrix $\mathbf{P}_k$ in Eq. (5) are no longer tightly capturing the error,

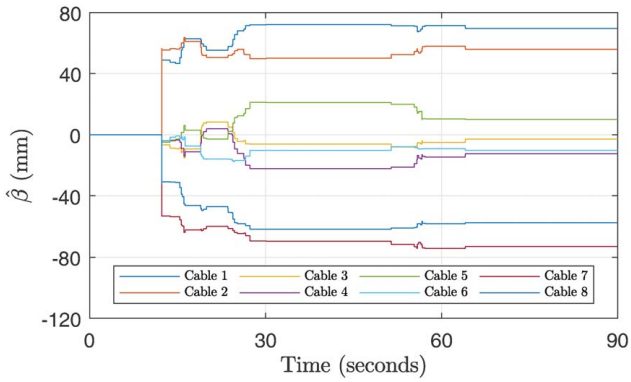


Fig. 12 Plots of the estimated cable-length biases from the experiments of Sec. 5

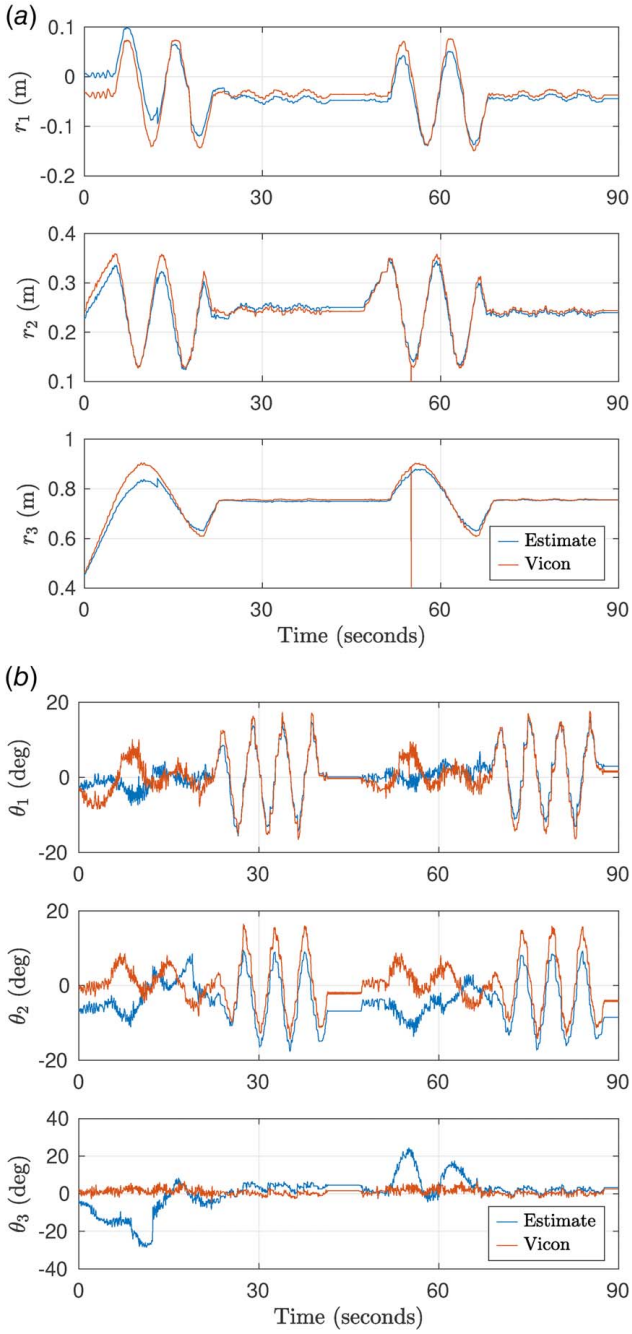


Fig. 13 Plots of the (*a*) position and (*b*) orientation estimates with the self-calibration algorithm (labeled "Estimate") and the ground-truth data from the Vicon system (labeled "Vicon") in the experiments of Sec. 5, where $\mathbf{r} = [r_1\ r_2\ r_3]^T$ and $\boldsymbol{\theta} = [\theta_1\ \theta_2\ \theta_3]^T$

but this is to be expected as the elastic cables result in unmodeled dynamics that are not captured by the loop-closure equations or measurement noise model. These results demonstrate that the proposed algorithm is still able to perform well under these challenging circumstances.

5 Experimental Results on a 6-DOF CDPR With Cable-Length Biases

To further validate the proposed self-calibration algorithm, it is implemented experimentally on the eight-cable 6-DOF CDPR

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shown in Fig. 11. This CDPR has a configuration similar to the IPAnema 2, much like the CDPR used in the numerical simulations of Sec. 4. The outer frame of the CDPR has a height of 1.27 m, a width of 1.575 m, and a depth of 0.91 m. The CDPR is operated through MATLAB SIMULINK using QUANSER'S QUARC real-time control software at a sampling rate of 100 Hz. A task-space-based proportional-derivative feedback controller with the improved closed-form tension distribution algorithm from Ref. [9, pp. 94–95] is used to perform motion control of the payload and follow a desired trajectory. The CDPR's motors are equipped with CUI AMT102-V incremental encoders, which are used to generate the cable-length measurements within the proposed self-calibration algorithm. A Vicon motion capture system with 6 MX40 cameras is used to obtain ground-truth data for the pose of the CDPR's payload. Using vendor specifications, this ground-truth system is estimated to have a positioning error with a standard deviation of at least 0.1 mm [55]. From our experience with this system, this is likely an optimistic estimate of the system's accuracy. Further details on the hardware, electronics, and software of the CDPR are found in Refs. [23,56].

The proposed algorithm outlined in Sec. 4.1 is implemented using the loop-closure equations of the form in Eq. (10). The damping is chosen as $\eta = 10^4$ for the inner-loop forward kinematics and $\eta = 10^3$ for the outer-loop self-calibration. The stopping criteria tolerances are chosen as $\|\hat{\boldsymbol{\rho}}_k^{(j+1)} - \hat{\boldsymbol{\rho}}_k^{(j)}\|_2 < 10^{-9}$, $\|\hat{\mathbf{x}}_k^{(j+1)} - \hat{\mathbf{x}}_k^{(j)}\|_2 < 10^{-9}$, and $j \leq 30$. As noted in Sec. 4.2, the norms $\|\hat{\boldsymbol{\rho}}_k^{(j+1)} - \hat{\boldsymbol{\rho}}_k^{(j)}\|_2$ and $\|\hat{\mathbf{x}}_k^{(j+1)} - \hat{\mathbf{x}}_k^{(j)}\|_2$ do not have a well-defined physical meaning or units. In this study, these stopping criteria are viewed from a purely numerical perspective and future work will examine the effect of choosing more physically meaningful criteria. The standard deviation of the cable-length measurement noise is approximated as 1 mm in the algorithm. The initial pose estimate used in the Levenberg–Marquardt algorithm is the pose estimate from the previous time-step. The threshold values of PDOP and ODOP for the outer loop are chosen as 20 mm and 10 deg, respectively. The number of time-steps to include in the outer-loop computations is chosen as $N = 20$. For this experiment, the self-calibration algorithm is run offline and computed in postprocessing, as if it was computed sequentially in time onboard the CDPR.

The estimated cable-length biases are plotted versus time in Fig. 12. Plots of the pose estimate versus time using the proposed self-calibration algorithm are shown in Fig. 13, along with ground-truth pose data measured from the Vicon system. Figure 14 includes the pose estimation error obtained with the proposed algorithm compared to the pose estimation error with only forward kinematics (i.e., without any bias estimation or self-calibration), as well as the $\pm 3\sigma$ bounds computed from the covariance matrix $\mathbf{P}_k$ in Eq. (5). Figure 12 demonstrates how roughly 12 s of data collection is required before the PDOP and ODOP thresholds are met and self-calibration begins. The cable-length biases converge to their steady-state values about 50 s into the experiment. It is shown in Fig. 13 that the pose estimate converges fairly well to the ground-truth pose values from Vicon, especially after the biases converge. This is further illustrated in Fig. 14, where large initial and transient pose errors are present due to the unknown biases, but then the pose errors converge mostly within the computed $\pm 3\sigma$ bounds. Figure 14 also shows that the pose estimation error with the self-calibration algorithm is significantly reduced compared to the use of forward kinematics alone once the cable-length biases converge sufficiently. This indicates that the self-calibration algorithm operates as intended and is able to effectively reduce the pose estimation errors induced by cable-length biases. It is also worth noting that the accuracy of the Vicon ground-truth data itself could be contributing to the fact that some of the errors with self-calibration fall slightly outside of the $\pm 3\sigma$ bounds.

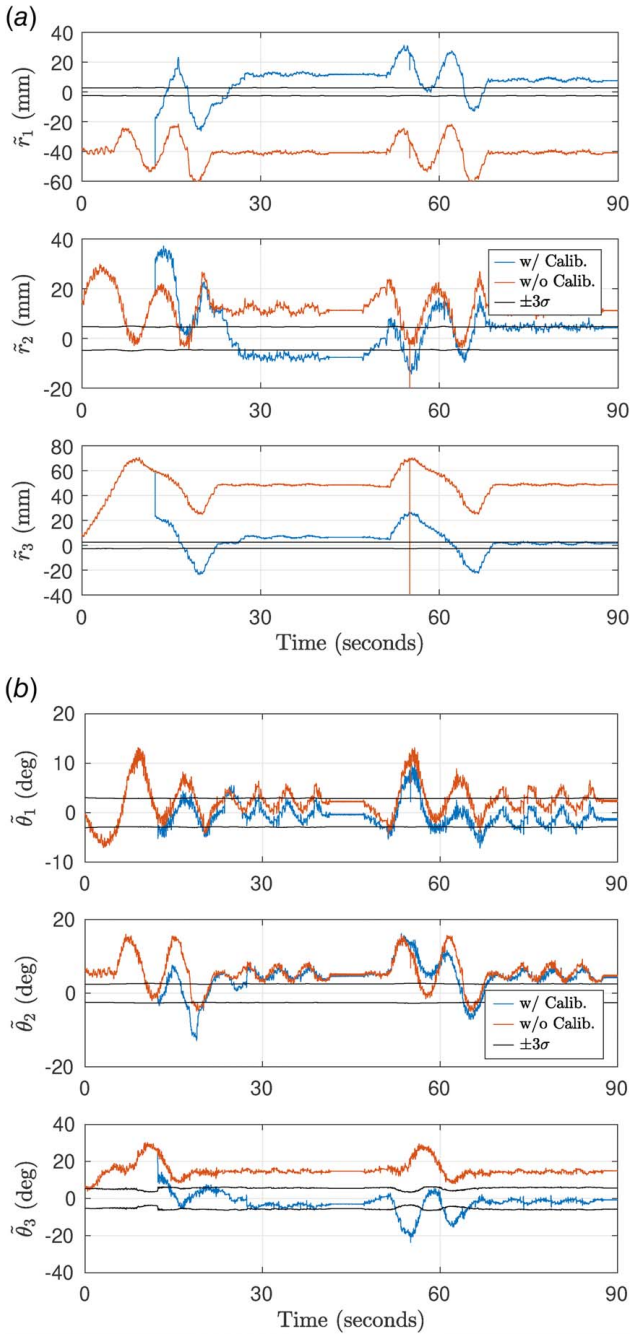


Fig. 14 Plots of the (*a*) position and (*b*) orientation estimation error with the self-calibration algorithm (labeled "w/ Calib.") and with only forward kinematics (labeled "w/o Calib.") in the experiments of Sec. 5, where $\mathbf{r} - \hat{\mathbf{r}} = [\tilde{r}_1 \ \tilde{r}_2 \ \tilde{r}_3]^T$ and $\boldsymbol{\theta} - \hat{\boldsymbol{\theta}} = [\tilde{\theta}_1 \ \tilde{\theta}_2 \ \tilde{\theta}_3]^T$

6 Conclusion

Novel PDOP and ODOP metrics inspired by the GPS navigation community were introduced in this article, which form the basis of the proposed self-calibration algorithm. These metrics are based on the covariance of the pose and bias estimation error, which makes them physically meaningful and intuitive to use in comparison to other metrics, such as observability indices. It was also shown through numerical simulations that using the PDOP and ODOP to carefully choose a limited number of rich data points for self-calibration results in smaller pose estimation error compared to the use of observability indices for this purpose. These results make a strong case for the use of PDOP and ODOP as metrics for data quality assessment when performing self-calibration.

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Experimental results with the proposed metrics and self-calibration algorithm confirm that they are effective in a real-world setting.

Future work on this topic will investigate applications of the proposed self-calibration algorithm in situations where more complicated measurement biases are present, such as CDPR geometry errors and pose-dependent or time-dependent biases.

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Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

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